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SYSTEM AND METHOD FOR LANDING RETRIEVABLE PROCESS MODULE

Abstract

A system includes a landing system configured to land a retrievable module on a subsea structure. The landing system includes a landing post, and a landing adjuster configured to adjust a position of the landing post relative to a landing side of the retrievable module.

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Background/Summary

BACKGROUND

[0001] The present disclosure generally relates to a landing system for a retrievable process module

used in subsea systems.

[0002] This section is intended to introduce the reader to various aspects of art that may be related to various aspects of the present disclosure, which are described and/or claimed below. This discussion is believed to be helpful in providing the reader with background information to facilitate a better understanding of the various aspects of the present disclosure. Accordingly, it may be understood that these statements are to be read in this light, and not as admissions of prior art.

[0003] In subsea applications, a retrievable module (e.g., retrievable process module, retrievable flow module, etc.) may be added to or retrieved from one or more subsea structures of a subsea system. The retrievable module includes a reconfigurable interior that may be altered based on the process to be performed by the retrievable module. However, due to the reconfiguration of internal components, a center of gravity of the retrievable module may be changed, which may inadvertently induce a bending moment on the retrievable module when placed on the subsea structure. Accordingly, a system for offsetting this induced bending moment is desired.

SUMMARY

[0004] A summary of certain embodiments disclosed herein is set forth below. It should be understood that these aspects are presented merely to provide the reader with a brief summary of these certain embodiments and that these aspects are not intended to limit the scope of this disclosure. Indeed, this disclosure may encompass a variety of aspects that may not be set forth below.

[0005] In certain embodiments, a system includes a landing system configured to land a retrievable module on a subsea structure. The landing system includes a landing post, and a landing adjuster configured to adjust a position of the landing post relative to a landing side of the retrievable module.

[0006] In certain embodiments, a system includes a retrievable module and a landing system configured to land the retrievable module on a subsea structure. The landing system includes a landing post including one or more compensators configured to move in response to a landing force, and a landing adjuster configured to adjust a position of the landing post relative to a landing side of the retrievable module.

[0007] In certain embodiments, a method including landing a retrievable module on a subsea structure via a landing system, including moving one or more compensators of a landing post in response to a landing force, and adjusting a position of the landing post relative to a landing side of the retrievable module via a landing adjuster.

[0008] Various refinements of the features noted above may exist in relation to various aspects of the present disclosure. Further features may also be incorporated in these various aspects as well. These refinements and additional features may exist individually or in any combination. For instance, various features discussed below in relation to one or more of the illustrated embodiments may be incorporated into any of the above-described aspects of the present disclosure alone or in any combination. The brief summary presented above is intended only to familiarize the reader with certain aspects and contexts of embodiments of the present disclosure without limitation to the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] These and other features, aspects, and advantages of the present disclosure will become better understood when the following detailed description is read with reference to the accompanying drawings in which like characters represent like parts throughout the drawings, wherein:

[0010] FIG. 1 is a schematic view of a subsea production system having at least one subsea

structure that includes a retrievable module, according to an embodiment of the present disclosure; [0011] FIG. 2 is a schematic view of one or more landing systems facilitating a soft landing of a retrievable module of FIG. 1 having varying mass distributions, according to an embodiment of the present disclosure;

[0012] FIG. 3 is a cross-sectional view of the landing system of FIG. 2, according to an embodiment of the present disclosure;

[0013] FIG. 4 is a schematic cross-sectional view of a landing sequence of the landing system of FIG. 2 having a low loading configuration, according to an embodiment of the present disclosure; and

[0014] FIG. 5 is a schematic cross-sectional view of a landing sequence of the landing system of FIG. 2 having a high loading configuration, according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0015] Certain embodiments commensurate in scope with the present disclosure are summarized below. These embodiments are not intended to limit the scope of the disclosure, but rather these embodiments are intended only to provide a brief summary of certain disclosed embodiments. Indeed, the present disclosure may encompass a variety of forms that may be similar to or different from the embodiments set forth below.

[0016] As used herein, the term “coupled” or “coupled to” may indicate establishing either a direct or indirect connection (e.g., where the connection may not include or include intermediate or intervening components between those coupled), and is not limited to either unless expressly referenced as such. The term “set” may refer to one or more items. Wherever possible, like or identical reference numerals are used in the figures to identify common or the same elements. The figures are not necessarily to scale and certain features and certain views of the figures may be shown exaggerated in scale for purposes of clarification.

[0017] Furthermore, when introducing elements of various embodiments of the present disclosure, the articles “a,” “an,” and “the” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Additionally, it should be understood that references to “one embodiment,” “an embodiment,” or “some embodiments” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Furthermore, the phrase A “based on” B is intended to mean that A is at least partially based on B. Moreover, unless expressly stated otherwise, the term “or” is intended to be inclusive (e.g., logical OR) and not exclusive (e.g., logical XOR). In other words, the phrase A “or” B is intended to mean A, B, or both A and B.

[0018] The present disclosure is generally directed toward a landing system (e.g., soft landing system, spring system, etc.) for a retrievable module used in subsea applications (e.g., below or under the surface of a body of water, such as the sea, ocean, lake river, or the like). As discussed above, the landing system disclosed herein is adjustable to a retrievable module of variable mass density. The landing system is configured to couple to the retrievable module (e.g., retrievable module hub) and make initial contact with a subsea structure, base, or floor. During a landing of the retrievable module onto a subsea structure, the landing system is compressed in multiple (e.g., three) separate stages to enable retrievable modules of varying mass densities to land onto the subsea structure. The first two compression stages of the landing system (e.g., the soft landing) include contacting the subsea structure and compression of a first compressible portion (e.g., a damper) so as to maintain contact with the subsea structure. The third compression stage involves a mass of the retrievable module causing compression of a second compressible portion (e.g., a spring), such that the second compressible portion bears the mass of the retrievable module (e.g., localized mass).

[0019] As such, in certain embodiments of the present disclosure, the landing system includes a landing system that compresses in multiple stages. By the end of a first compression stage, both

inner and outer extension nuts disposed on a bottom side of the landing system contact the subsea structure. During a second compression stage, a resistive pressure is applied by the first compressible portion to provide a soft-landing phase after initially making contact. During a third compression stage, the load of the retrievable module is placed on the second compressible portion. [0020] In certain embodiments of the present disclosure, the landing system includes the inner and outer extension nuts disposed on the axial end of the landing system that contacts the subsea structure. The inner and outer extension nuts are axially coupled to the landing system and are configured to elongate the landing system to compensate for increased and/or decreased compression of the first and/or second compressible portions due to variance in the load distribution of the retrievable module.

[0021] With the foregoing in mind, FIG. 1 is a schematic view of a subsea system **10** having various subsea equipment that may receive the retrievable module using the landing system described in further detail below. As illustrated, the subsea system includes electrical cables **12** used for transmitting information and primary electrical power for various subsea components (e.g., actuators, sensors, etc.). The subsea system **10** may include a subsea hydrocarbon production system configured to extract oil or gas from a subterranean reservoir, a subsea fluid injection system configured to inject fluid (e.g., liquid or gas) into a subterranean reservoir, or any other subsea system associated with subterranean reservoirs. In certain embodiments, the subsea system **10** may include a subsea tree **14** coupled to a wellhead **16** to form a subsea station **18** configured to extract and/or inject fluids relative to a subterranean reservoir. For example, the subsea station **18** may be configured to extract formation fluid, such as oil and/or natural gas, from the sea floor **20** through the well **22**. In some embodiments, the subsea system **10** may include multiple subsea stations **18** that extract and/or inject fluids relative to respective wells **22**.

[0022] In embodiments of the subsea system **10** configured for production, after passing through the subsea tree **14**, the formation fluid flows through fluid conduits or pipes **24** to a pipeline manifold **26**. The pipeline manifold **26** may connect to one or more flowlines **28** to enable the formation fluid to flow from the wells **22** to a surface platform **30**. In some embodiments, the surface platform **30** may include a floating production, storage, and offloading unit (FPSO) or a shore-based facility. In addition to flowlines **28** that carry the formation fluid away from the wells **22**, the subsea system **10** may include lines or conduits **32** that supply fluids, as well as carry control and data lines to the subsea equipment. These conduits **32** connect to a distribution module **34**, which in turn couples to the subsea stations **18** via supply lines **36**. In some scenarios, the platform **30** may be located a significant distance (e.g., greater than 100 m, greater than 1 km, greater than 10 km, or greater than 60 km) away from the wells **22**. As discussed in further detail below, the subsea system **10** (e.g., the subsea tree **14**, the subsea station **18**, the pipeline manifold **26**, and/or the distribution module **34**) may include a connection or docking area, wherein a retrievable module **38** may be configured to land and connect with the subsea system **10**. While the subsea system **10** described above is for extracting hydrocarbons, it should be understood that the present disclosure may also apply to other types of subsea systems **10** such as subsea injection systems (e.g., subsea gas injection system, subsea water injection system, and/or subsea carbon dioxide injection system).

[0023] In certain embodiments, at least one of the subsea stations **18** of the subsea system **10** may include the retrievable module **38** that may be retrievable from the subsea station **18**. The retrievable module **38** may be configured to attach to the tree **14** or, in certain embodiments, other structures of the subsea system **10**, such as the pipeline manifold **26** and/or the distribution module **34**. As discussed in further detail herein, the retrievable module **38** may include one or more landing systems **40** to assist in orienting the retrievable module **38** relative to the subsea station **18** and making various connections, such as fluid connections, electrical power connections, communication and control connections, or any combination thereof. It may be recognized that the subsea system **10** disclosed herein is just an example of many possible arrangements. The

retrievable module **38** and the landing systems **40** may be compatible with any number of arrangements of the subsea system **10**. For example, the subsea system **10** may or may not include the platform **30**.

[0024] FIG. **2** is a schematic view of one or more landing systems **40** facilitating a soft landing of a retrievable module **38** having varying mass distributions. In the illustrated embodiment, the retrievable module **38** includes components **60** having a center of gravity **62** disposed within an interior **64** of the retrievable module **38**. As shown, the retrievable module **38** includes an arm portion **66** that is designed to engage a capture envelope **68** (e.g., a docketing area or volume having various connections, such as mechanical connections, electrical connections, fluid connections, etc.) of a subsea structure **70** (e.g., subsea station, subsea tree, subsea manifold etc.). The retrievable module **38** may include a reusable exterior **71** (e.g., housing, body, or framework) that may be outfitted in the interior **64** with components **60** of varying masses, thereby varying the mass distribution of the retrievable module **38**. In certain embodiments, the components **60** may include one or more valves (e.g., gate valves, ball valves, check valves, chokes, blowout preventers, etc.), pumps, compressors, flowmeters, sensors (e.g., pressures sensors, temperature sensors, etc.), chemical injection systems (e.g., chemical injection metering valves), fluid intervention systems, subsea sampling systems, tree conversion systems (e.g., producer to injector), high-integrity pressure protection systems (HIPPS), fluid conduits, fluid tanks, accumulators, actuators, chokes, controllers (e.g., processor-based controllers having one or more processors, memory, and instructions to control the components), or any combination thereof.

[0025] As shown in view **72**, a first retrievable module **74** includes first components **76** having a first center of gravity **78** at a first location **80**. The first retrievable module **74** also includes a first arm portion **81**. The first retrievable module **74** is oriented at a first orientation **82** (e.g., pose) relative to the subsea structure **70**. In view **84**, a second retrievable module **86** includes second components **88** having a second center of gravity **90** at a second location **92**. The second retrievable module **86** also includes a second arm portion **93**. As shown, the second retrievable module **86** has a second orientation **94** relative to the subsea structure **70**. In the illustrated embodiment, the first orientation **82** and the second orientation **94** are different from one another due to the variance in mass distribution (e.g., differences between the first center of gravity **78** and the second center of gravity **90**) of the first components **76** and the second components **88**. As shown in the views **72** and **84**, neither the first arm portion **81** of the first retrievable module **74** nor the second arm portion **93** of the second retrievable module **86** are positioned within the capture envelope **68** of the subsea structure **70**. Although the illustrated embodiment shows the retrievable module **38** as having a certain shape, it may be recognized that the retrievable module **38** may have a variety of shapes.

[0026] As shown in view **96**, the first retrievable module **74** is outfitted with a landing system **40** having landing posts **97** (e.g., landing posts **98**, **100**) on a bottom side **101** (e.g., landing side) of the retrievable module **74**. Additionally, in view **102**, the second retrievable module **86** is outfitted with the landing system **40** having the landing posts **97** (e.g., landing posts **104** and **106**) on the bottom side **101** of the retrievable module **86**. Both the landing posts **98**, **100** and the landing posts **104**, **106** respectively have the same positions with respect to the retrievable module **38**. As shown, due to both the first retrievable module **74** and the second retrievable module **86** having the landing systems **40**, both the first arm portion **81** of the first retrievable module **74** and the second arm portion **93** of the second retrievable module **86** are positioned within the capture envelope **68** of the subsea structure **70**. Because both the first arm portion **81** and the second arm portion **93** are aligned within the capture envelope **68** despite the first retrievable module **74** and the second retrievable module **86** having centers of gravity **62** at different locations (e.g., different mass distributions), it may be appreciated that the landing systems **40** may align (e.g., stabilize and orient an angle, a height, and a position of) retrievable modules **38** with different mass distributions with respect to the capture envelope **68**. In this manner, the landing systems **40** enable a successful

landing (e.g., capture) of the retrievable module **38** on the subsea structure **70** irrespective of the center of gravity **62** (e.g., mass distribution) of the components **60** (e.g., internal components) of the retrievable module **38**, thereby saving time and cost of having to redesign the retrievable module **38**.

[0027] FIG. **3** is a cross-sectional view of the landing system **40**. The landing system **40** may be described with respect to an axial direction **120** (e.g., axial axis), a radial direction **122** (e.g., radial direction), and a circumferential direction **124** (e.g., circumferential axis). In the illustrated embodiment, the landing system **40** includes an outer sleeve **130** disposed about an inner sleeve **132**. As shown, the outer sleeve **130** includes a bottom section **131**, a middle section **133** and a top section **134**. The outer sleeve **130** is configured to slide past (e.g., slide over, telescope past) the inner sleeve **132**. In other words, the outer and inner sleeves **130** and **132** may be coaxial or concentric sleeves that axially expand and contract relative to one another in the axial direction **120**. Additionally, the landing system **40** includes a stop pin **136** (e.g., hollow shaft portion or soft-landing piston) and a first compressible portion **138** (e.g., compensator, fluid damper) disposed in a first interior **142** (e.g., chamber) of the outer sleeve **130**. The landing system **40** also includes an axial core portion **144** (e.g., central shaft portion), a second compressible portion **146** (e.g., compensator, spring), and a stopper **148** (e.g., annular stopper) disposed in a second interior **150** of the inner sleeve **130**. The second compressible portion **146** may include one or more springs, such as one or more mechanical springs (e.g., helical springs). In some embodiments, the second compressible portion **146** may include one or more fluid springs (e.g., pneumatic springs or shock absorbers) and/or one or more magnetic springs (e.g., magnetic compensator). In some embodiments, the second compressible portion **146** may include one or more mechanical springs, one or more fluid springs, one or more magnetic springs, or a combination thereof. As shown, the axial core portion **144**, the second compressible portion **146**, and the stopper **148** may also be at least partially disposed in the first interior **142** of the outer sleeve **130**. The landing system **40** also includes an outer extension nut **151** (e.g., threaded nut, landing adjuster) coupled to the inner sleeve **132**, and an inner extension nut **153** (e.g., threaded nut, landing adjuster) coupled to the axial core portion **144**.

[0028] In the illustrated embodiment, the axial core portion **144** is disposed axially adjacent to a first axial end **152** of the stop pin **136**. Additionally, the first compressible portion **138** is disposed axially adjacent to a second axial end **154** of the stop pin **136**. As shown, the stopper **148** is disposed circumferentially about both the axial core portion **144** and the stop pin **136** and is able to axially slide past both the axial core portion **144** and the stop pin **136**. The stopper **148** includes a protrusion **156** (e.g., annular protrusion or flange) from an outer surface **158** of the stopper **148**. The protrusion **156** is configured to abut and compress a free axial end **160** of the second compressible portion **146**.

[0029] As shown, the second compressible portion **146** is disposed about the axial core portion **144** and also about a lower stopper portion **162** of the stopper **148**. A fixed axial end **164** of the second compressible portion **146** is coupled to an interior axial surface **166** of the inner sleeve **132**. The second compressible portion **146** (e.g., spring) is configured to be compressed by the stopper **148** into the second interior **150** of the inner sleeve **132**. Although the illustrated embodiment shows the second compressible portion **146** as being a spring, it may be recognized that the second compressible portion **146** may include other compressible devices (e.g., hydraulics, dampers, etc.).

[0030] The first compressible portion **138** is disposed with the first interior **142** of the outer sleeve **130** and is configured to be compressed by the stop pin **136** (e.g., soft-landing piston). In certain embodiments, the first compressible portion **138** is less stiff than the second compressible portion **146**. In the illustrated embodiment, the first compressible portion **138** is illustrated as being a damper (e.g., water-filled damper), however it may be recognized that the first compressible portion **138** may include other compressible devices (e.g., hydraulics, springs, etc.). The first compressible portion **138** also may include one or more fluid ports or openings to facilitate a

controlled discharge of fluid (e.g., water) during a soft landing when the stop pin **136** (e.g., soft-landing piston) moves upwardly in the first compressible portion **138**, while intaking fluid (e.g., water) when not soft landing.

[0031] As shown, a bottom flange **168** of the bottom section **131** of the outer sleeve **130** is coupled to a middle flange **170** of the middle section **133** of the outer sleeve **130** via fasteners (e.g., threaded fasteners, screws, or bolts) passing through a plurality of holes **172** (e.g., bolt holes). In certain embodiments, the plurality of holes **172** (e.g., holes **174** and **176**) may enable a coupling the bottom section **131** and/or the middle section **133** of the outer sleeve **130** to the retrievable module, thereby coupling the landing system **40** to the retrievable module. Additionally, the middle section **133** of the outer sleeve **130** is configured to couple to the top section **134** of the outer sleeve **130** via fasteners (e.g., threaded fasteners, screws, or bolts) passing through a plurality of holes **178** (e.g., holes **180**, **182**).

[0032] In the illustrated embodiment, the outer extension nut **151** is coupled to a sleeve axial end **184** of the inner sleeve **132**, and the inner extension nut **153** coupled to a core axial end **186** of the axial core portion **144**. The outer extension nut **151** is configured to axially slide relative to the inner sleeve **132**, and the inner extension nut **153** is configured to axially slide relative to the axial core portion **144**. In certain embodiments, the outer extension nut **151** may be threaded onto the sleeve axial end **184** of the inner sleeve **132**, and the inner extension nut **153** may be threaded onto the core axial end **186** of the axial core portion **144**. An outer lock nut **188** may be disposed between the outer extension nut **151** and the inner sleeve **132** to lock a position of the outer extension nut **151** relative to the inner sleeve **132**. Additionally, an inner lock nut **190** may be disposed between the inner extension nut **153** to lock a position of the inner extension nut **153** relative to the axial core portion **144**. In certain embodiments, a specialized tool (e.g., specialized wrench) may be used to adjust the outer extension nut **151** and/or the inner extension nut **153**. The outer extension nut **151** and/or the inner extension are configured to contact the subsea structure during the landing of the retrievable module **38**.

[0033] In the illustrated embodiment, the bottom portion **131** of the outer sleeve **130** includes an inward protrusion **190** (e.g., inner annular protrusion) configured to interlock with an outward protrusion **192** (e.g., outer annular protrusion) disposed on an outer surface **194** of the inner sleeve **132**. As shown, the outward protrusion **192** and the inward protrusion **190** are configured to axially block the inner sleeve **132** from continuing to slide axially relative to the outer sleeve **130**, so that the inner sleeve **132** remains axially secured or captured within the outer sleeve **130**.

[0034] In the illustrated embodiment, the axial core portion **144** includes an outward core protrusion **196** (e.g., outer annular protrusion) and the stopper **140** includes an inward stopper protrusion **198** (e.g., inner annular protrusion). As shown, the outward core protrusion **196** and the inward stopper protrusion **198** are configured to axially block the stopper **140** from axially sliding off the axial core portion **144**. Additionally, the protrusion **156** of the stopper **140** axially abuts the second compressible portion **146**, ensuring that the stopper **140** remains disposed over the interface **200** (e.g., axial abutment) between the axial core portion **144** and the stop pin **136**.

[0035] In the illustrated embodiment, the outer sleeve **130**, the inner sleeve **132**, the stop pin **136**, the first compressible portion **138**, the stopper **140**, the axial core portion **144**, the second compressible portion **146**, the outer extension nut **151**, and the inner extension nut **153** are shown as being annular in shape and disposed about a central axis **202**. In certain embodiments, the outer sleeve **130**, the inner sleeve **132**, the stop pin **136**, the first compressible portion **138**, the stopper **140**, the axial core portion **144**, the second compressible portion **146**, the outer extension nut **151**, the inner extension nut **153**, or a combination thereof, may not be annular in shape and/or may be radially offset from the central axis **202**.

[0036] FIG. **4** is a schematic cross-sectional view of a landing sequence **230** of the landing system **40** while supporting a retrievable module **38** (e.g., portion of a retrievable module **38**) having a low mass (e.g., low mass density). In the illustrated embodiment, the landing system **40** is directly

coupled to the retrievable module **38** (e.g., retrievable module hub) and is being lowered onto the subsea structure **70**. As discussed herein, the subsea structure **70** may include a subsea tree, or any other subsea structure used in a subsea system. As shown, the landing sequence **230** includes views **232**, **234**, **236**, **238**, **240**, and **242**, which represent the landing system **40** chronologically during a landing of the retrievable module **38** onto the subsea structure **70**. It may be recognized that these views may occur in the reverse order during a retrieval of the retrievable module **38** from the subsea structure **70**. Each view includes three dimensions, including a first length **246** of the first compressible portion **138**, a distance **248** spanning from a free axial end **250** of the stopper **148** and an interior axial surface **252** of the middle portion **132** of the outer sleeve **130**, and a second length **254** of the second compressible portion **146**.

[0037] Additionally, each of the views includes an overhang distance **256** between an axial end **258** of the outer extension nut **151** and the inner sleeve **132**. In certain embodiments, the overhang distance **256** between the axial end **258** of the outer extension nut **151** and the inner sleeve **132** is equivalent to an overhang distance **260** between the inner extension nut **153** and the axial core portion **144**, since the outer extension nut **151** and the inner extension nut **153** may be adjusted by the same axial distance. Because the outer extension nut **151** does not slide relative to the inner sleeve **132** during the landing, the overhang distance **256** remains the same during the landing sequence **230**. The same holds true for the overhang distance **260**.

[0038] In the view **232**, the landing system **40** coupled to the retrievable module **38** is approaching the subsea structure **70** in a vertical downward direction. As discussed in more detail herein, because the retrievable module **38** has a minimal (e.g., low) mass density in the landing sequence **230**, the overhang distance **256** (e.g., and overhang distance **260**) are set to a low value. In the view **232**, neither the first compressible portion **138** nor the second compressible portion **146** are compressed.

[0039] In the view **234**, the inner extension nut **153** contacts the subsea structure **70**. The first length **246** of the first compressible portion **138**, the distance **248** spanning from a free axial end **250** of the stopper **148** and the interior axial surface **252** of the bottom portion **131** of the outer sleeve **130**, and the second length **254** of the second compressible portion **146** have not changed from the view **232**. The inner extension nut **153** contacting the subsea structure **70** marks the beginning of a first compression stage **262** of the landing system **240**. During the first compression stage **262**, the axial core portion **144** drives the stop pin **136** (e.g., soft-landing piston) to compress and/or discharge fluid (e.g., water) in the first compressible portion **138**. Additionally, the inner sleeve **132** simultaneously slides axially past the axial core portion **144** in the direction **264**, wherein the direction **264** is opposite of the axial direction **120**.

[0040] In the view **236**, the both the inner extension nut **153** and the outer extension nut **151** are in contact with the subsea structure **70**, marking the end of the first compression stage **262** and the beginning of the second compression stage **266**. As shown, by the end of the first compression stage **262**, the first length **246** of the first compressible portion **138** has decreased due to compression of the first compressible portion **138**. However, the distance **248** spanning from the free axial end **250** of the stopper **148** and the interior axial surface **252** of the bottom portion **131** of the outer sleeve **130**, and the second length **254** of the second compressible portion **146** have not changed from the view **234**.

[0041] In the view **238**, the landing system **40** is shown as undergoing the second compression stage **266**. During the second compression stage **266**, the axial core portion **144** continues to drive the stop pin **136** to compress and/or discharge fluid (e.g., water) in the first compressible portion **138**. Additionally, during the second compression stage **266**, the outer sleeve **130** is configured simultaneously to slide toward (but not yet contact) the stopper **148** due to the outer sleeve **130** sliding over the inner sleeve **132**. As shown, during the second compression stage **266**, the first length **246** of the first compressible portion **138** has continued to decrease due to compression of the first compressible portion **138**. Additionally, the distance **248** spanning from the free axial end

250 of the stopper **148** and the interior axial surface **252** has decreased due to outer sleeve **130** sliding over the inner sleeve **132**. However, the second length **254** of the second compressible portion **146** has not changed from the view **236**. The second compressible stage **266** may also be known as a soft-landing stage of the landing system **40**.

[0042] In the view **240**, the outer sleeve **130** contacts the stopper **148**, thereby marking an end of the second compression stage **266** and a beginning of the third compression stage **268**. During the third compression stage **268**, the axial core portion **144** continues to compress the first compressible portion **138**. Additionally, during the third compression stage **268**, the outer sleeve **130** simultaneously drives the stopper **148** to compress the second compressible portion **146** (e.g., spring) via the protrusion **156** of the stopper **148**. As shown, by the beginning of the third compression stage **268**, the first length **246** of the first compressible portion **138** has continued to decrease due to compression of the first compressible portion **138**. Additionally, the distance **248** spanning from the free axial end **250** of the stopper **148** and the interior axial surface **252** is now zero due to contact between the interior axial surface **252** and the stopper **148**. However, the second length **254** of the second compressible portion **146** (e.g., spring) has not changed from the view **238**.

[0043] In the view **242**, the outer sleeve **130**, which is coupled to the retrievable module **38**, continues to drive the stopper **148** to compress the second compressible portion **146** (e.g., spring). As shown, the outer sleeve **130** has also continued to slide over (e.g., past, relative to) the inner sleeve **132**. The axial core portion **144** has simultaneously continued to compress the first compressible portion **138**. As shown, by the end of the third compression stage **268**, the first length **246** of the first compressible portion **138** has continued to decrease due to compression of the first compressible portion **138**. Additionally, the distance **248** spanning from the free axial end **250** of the stopper **148** and the interior axial surface **252** remains zero due to contact between the interior axial surface **252** and the stopper **148**. Furthermore, the second length **254** of the second compressible portion **146** (e.g., spring) has decreased due to compression of the second compressible portion **146**.

[0044] At the end of the third compression stage **268**, the retrievable module **38** is separated by a distance **270** from the subsea structure **70**. Additionally, at the end of the third compression stage **268** (as shown in the view **242**), the interior axial surface **250** of the outer sleeve **130** is separated from the interior axial surface **166** of the inner sleeve **132** by a distance **272**. As shown, between the view **232** and the view **242**, the inner sleeve **132**, the stop pin **136**, the first compressible portion **138**, the axial core portion **144**, the second compressible portion **146**, and the stopper **148** have moved relative to the outer sleeve **130**, which is fixed to the retrievable module **38**.

[0045] FIG. 5 is a schematic cross-sectional view of a landing sequence **290** of the landing system **40** while supporting a retrievable module **38** (e.g., portion of a retrievable module **38**) having a higher mass (e.g., higher mass density) compared to the landing sequence shown in FIG. 4. In the illustrated embodiment, the landing system **40** is directly coupled to the retrievable module **38** (e.g., via fasteners) and is being lowered onto the subsea structure **70**. As discussed herein, the subsea structure **70** may include a subsea tree, or any other subsea structure used in a subsea system. As shown, the landing sequence **290** includes views **292**, **294**, **296**, **298**, **300**, and **302**, which represent the landing system **40** chronologically during a landing of the retrievable module **38** onto the subsea structure **70**. It may be recognized that these views may occur in the reverse order during a retrieval of the retrievable module **38** from the subsea structure **70**. Each view includes three dimensions, including a first length **306** of the first compressible portion **138**, a distance **308** spanning from a free axial end **250** of the stopper **148** and an interior axial surface **252** of the middle portion **132** of the outer sleeve **130**, and a second length **314** of the second compressible portion **146**.

[0046] Additionally, the view **292** includes an overhang distance **316** between the axial end **258** of the outer extension nut **151** and the inner sleeve **132**. In certain embodiments, the overhang

distance **316** between the axial end **258** of the outer extension nut **151** and the inner sleeve **132** is equivalent to an overhang distance **320** between the inner extension nut **153** and the axial core portion **144**, since the outer extension nut **151** and the inner extension nut **153** may be adjusted by the same axial distance. Because the outer extension nut **151** does not slide relative to the inner sleeve **132** during the landing, the overhang distance **316** remains the same during the landing sequence **290**. The same holds true for the overhang distance **320**. As shown, the overhang distances **316** and **320** in the landing sequence **290** are greater than the respective overhang distances shown in the landing sequence **230**. As discussed further herein, the increase in the overhang distances **316** and **320** is to compensate for the increased mass of the retrievable module **38**.

[0047] In the view **292**, due to the increased overhang distances **316** and **320**, the inner extension nut **153** is already in contact with the subsea structure **70**. The inner extension nut **153** contacting the subsea structure **70** marks the beginning of the first compression stage **262** of the landing system **40**. During the first compression stage **262**, the axial core portion **144** drives the stop pin **136** (e.g., soft-landing piston) to compress and/or discharge fluid (e.g., water) in the first compressible portion **138**. Additionally, the inner sleeve **132** simultaneously slides axially past the axial core portion **144** in the direction **264**, wherein the direction **264** is opposite of the axial direction **120**.

[0048] In the view **294**, both the inner extension nut **153** and the outer extension nut **151** are in contact with the subsea structure **70**, marking the end of the first compression stage **262** and the beginning of the second compression stage **266**. As shown, by the end of the first compression stage **262**, the first length **306** of the first compressible portion **138** has decreased due to compression of the first compressible portion **138**. However, the distance **308** spanning from the free axial end **250** of the stopper **148** and the interior axial surface **252** of the bottom portion **131** of the outer sleeve **130**, and the second length **314** of the second compressible portion **146** have not changed from the view **292**.

[0049] In the view **296**, the landing system **40** is shown as undergoing the second compression stage **266**. During the second compression stage **266**, the axial core portion **144** continues to drive the stop pin **136** to compress and/or discharge fluid (e.g., water) in the first compressible portion **138**. Additionally, during the second compression stage **266**, the outer sleeve **130** is configured to simultaneously slide toward (but not yet contact) the stopper **148** due to the outer sleeve **130** sliding over the inner sleeve **132**. As shown, during the second compression stage **266**, the first length **306** of the first compressible portion **138** has continued to decrease due to compression of the first compressible portion **138**. Additionally, the distance **308** spanning from the free axial end **250** of the stopper **148** and the interior axial surface **252** has decreased due to outer sleeve **130** sliding over the inner sleeve **132**. However, the second length **314** of the second compressible portion **146** has not changed from the view **294**. The second compressible stage **266** may also be known as a soft-landing stage of the landing system **40**.

[0050] In the view **298**, the outer sleeve **130** contacts the stopper **148**, thereby marking an end of the second compression stage **266** and a beginning of the third compression stage **268**. During the third compression stage **268**, the axial core portion **144** continues to compress the first compressible portion **138**. Additionally, during the third compression stage **268**, the outer sleeve **130** simultaneously drives the stopper **148** to compress the second compressible portion **146** (e.g., spring) via the protrusion **156** of the stopper **148**. As shown, by the beginning of the third compression stage **268**, the first length **306** of the first compressible portion **138** has continued to decrease due to compression of the first compressible portion **138**. Additionally, the distance **308** spanning from the free axial end **250** of the stopper **148** and the interior axial surface **252** is now zero due to contact between the interior axial surface **252** and the stopper **148**. However, the second length **314** of the second compressible portion **146** (e.g., spring) has not changed from the view **296**.

[0051] In the view **300**, the landing system **40** is shown as undergoing the third compression stage **268**. As shown, the outer sleeve **130**, which is coupled to the retrievable module **38**, continues to drive the stopper **148** to compress the second compressible portion **146** (e.g., spring). As shown, the outer sleeve **130** has also continued to slide over (e.g., past, relative to) the inner sleeve **132**. The axial core portion **144** has simultaneously continued to compress the first compressible portion **138**. As shown, the first length **306** of the first compressible portion **138** has continued to decrease due to compression of the first compressible portion **138**. Additionally, the distance **308** spanning from the free axial end **250** of the stopper **148** and the interior axial surface **252** remains zero due to contact between the interior axial surface **252** and the stopper **148**. Furthermore, the second length **314** of the second compressible portion **146** (e.g., spring) has decreased due to compression of the second compressible portion **146**.

[0052] In the view **302**, the outer sleeve **130** continues to drive the stopper **148** to compress the second compressible portion **146**. As shown, the outer sleeve **130** has also continued to slide over (e.g., past, relative to) the inner sleeve **132**. The axial core portion **144** has simultaneously continued to compress the first compressible portion **138**. As shown, by the end of the third compression stage **268**, the first length **306** of the first compressible portion **138** has continued to decrease due to compression of the first compressible portion **138**. Additionally, the distance **308** spanning from the free axial end **250** of the stopper **148** and the interior axial surface **252** remains zero due to contact between the interior axial surface **252** and the stopper **148**. Furthermore, the second length **314** of the second compressible portion **146** has decreased due to compression of the second compressible portion **146** (e.g., spring).

[0053] In the illustrated embodiment, in the view **302** (e.g., at the end of the third compression stage **268**), the first length **306** of the first compressible portion **138** is smaller than the first length **246** of the first compressible portion **138** shown in FIG. 4. Additionally, as shown, the second length **314** of the second compressible portion **146** is smaller than the second length **254** of the second compressible portion **146** shown in FIG. 4. That is, after completion of the third compression **268**, the lengths of the first compressible portion **138**, the second compressible portion **146**, or a combination thereof, are based on a mass of the load (e.g., the retrievable module **38**) supported by the landing system **40**. For example, in response to the retrievable module **38** having a higher mass density, the lengths of the first compressible portion **138**, the second compressible portion **146**, or the combination thereof, is configured to decrease. Additionally, at the end of the third compression stage **268** (as shown in the view **302**), the interior axial surface **250** of the outer sleeve **130** is separated from the interior axial surface **166** of the inner sleeve **132** by a distance **322**. As shown, compared to the distance **272** of FIG. 4, the distance **322** between the inner sleeve **132** and the outer sleeve **130** has decreased. As shown in the view **302**, the retrievable module **38** is separated by the distance **270** from the subsea structure **70** after completion of the third compression stage **268**.

[0054] As shown, both the overhang distance **316** of the outer extension nut **151** and the overhang distance **320** of the inner extension nut **153** are greater than the overhang distances **256** and **260** (shown in FIG. 4), respectively. In the illustrated embodiment, the amount of increase in overhang of the outer extension nut **151** and the inner extension nut **153** matches the amount of decrease in the distance **322** between the interior axial surfaces **250** and **166**. That is, the overhang distances **316** and **320** compensate for the decrease in the distance **322** (e.g., increase in overlap between the inner sleeve **132** and the outer sleeve **130**) due to the increase load of the retrievable module **38**. It may be appreciated that the outer extension nut **151** and/or the inner extension nut **153** are configured to adjust the position of the landing post **40** relative to the landing side **101** of the retrievable module **38** in a first crosswise direction (e.g., vertical direction **264**) to the landing side **101**. In certain embodiments, the outer extension nut **151** and/or the inner extension nut **153** may be configured to adjust the position of the landing post **40** relative to the landing side **101** of the retrievable module **38** in a second crosswise direction (e.g., direction **122**) to the landing side **101**.

That is, the outer extension nut **151** (e.g., landing adjuster) and/or the inner extension nut **153** (e.g., landing adjuster) may compensate (e.g., via the overhang of the extension nuts **151** and **153**) for the change in overlap (e.g., position) of the inner sleeve **132** and the outer sleeve **130** to stabilize landing of retrievable modules **38** of different masses, centers of gravity, or a combination thereof. [0055] Technical effects of the disclosed embodiments include a landing system designed to attach to a retrievable module and offset a moment induced onto a retrievable module due to an imbalance in mass. The landing system includes a soft-landing compressible portion for making initial contact and a load-bearing compressible portion that bears the load of the retrievable module. The load-bearing compressible portion and/or the soft-landing compressible portion may be displaced by varying amounts based on the mass of the retrievable module. To offset the displacement of either the load-bearing compressible portion or the soft-landing compressible portion, the landing system includes inner and outer extension nuts disposed on an axial end of the landing system that may be adjusted to compensate for the different compression distances. As such, the landing system may be adjusted based on the mass distribution of the retrievable module to ensure that retrievable modules of variable mass distribution may land on a subsea structure while avoiding an induced bending moment due to a mass imbalance, thereby avoiding the expenditure of time and money of having to redesign the retrievable module each time the mass distribution of the retrievable module is reconfigured.

[0056] The subject matter described in detail above may be defined by one or more clauses, as set forth below.

[0057] A system includes a landing system configured to land a retrievable module on a subsea structure. The landing system includes a landing post, and a landing adjuster configured to adjust a position of the landing post relative to a landing side of the retrievable module.

[0058] The system of the preceding clause, wherein the landing post includes one or more compensators configured to move in response to a landing force.

[0059] The system of any preceding clause, wherein the one or more compensators include one or more of a fluid compensator, a mechanical compensator, a magnetic compensator, or a combination thereof.

[0060] The system of any preceding clause, wherein the one or more compensators include a plurality of compensators configured to actuate in a plurality of stages.

[0061] The system of any preceding clause, wherein the plurality of compensators includes a fluid compensator including a piston disposed in a chamber and a mechanical compensator including a spring.

[0062] The system of any preceding clause, including a plurality of landing systems configured to land the retrievable module on the subsea structure, wherein each of the plurality of landing systems includes the landing post and the landing adjuster.

[0063] The system of any preceding clause, wherein the landing adjuster is configured to adjust the position to stabilize landing of different masses, centers of gravity, or a combination thereof, of the retrievable module.

[0064] The system of any preceding clause, wherein the landing adjuster is configured to adjust the position of the landing post relative to the landing side of the retrievable module in one or more directions relative to the landing side.

[0065] The system of any preceding clause, wherein the landing adjuster is configured to adjust the position of the landing post relative to the landing side of the retrievable module in a first direction crosswise to the landing side.

[0066] The system of any preceding clause, wherein the landing adjuster is configured to adjust the position of the landing post relative to the landing side of the retrievable module in a second direction along the landing side.

[0067] The system of any preceding clause, wherein the first direction is vertical, and the second direction is horizontal.

[0068] The system of any preceding clause, wherein the landing adjuster includes a manual landing adjuster, an automatic landing adjuster, or a combination thereof.

[0069] The system of any preceding clause, wherein the landing adjuster includes one or more threaded landing adjusters.

[0070] The system of any preceding clause, including the retrievable module including one or more of a valve, a pump, a compressor, a flowmeter, a sensor, a chemical injection system, a fluid intervention system, a subsea sampling system, a tree conversion system, a high-integrity pressure protection system, a fluid conduit, a fluid tank, an accumulator, an actuator, a choke, a controller, or any combination thereof.

[0071] The system of any preceding clause, including the subsea structure including a subsea tree, a subsea manifold, or a combination thereof.

[0072] A system including a retrievable module and a landing system configured to land the retrievable module on a subsea structure. The landing system includes a landing post including one or more compensators configured to move in response to a landing force, and a landing adjuster configured to adjust a position of the landing post relative to a landing side of the retrievable module.

[0073] The system of the preceding clause, wherein the landing adjuster is configured to adjust the position of the landing post relative to the landing side of the retrievable module in one or more directions relative to the landing side, wherein the landing adjuster is configured to adjust the position to stabilize landing of different masses, centers of gravity, or a combination thereof, of the retrievable module.

[0074] The system of any preceding clause, wherein the one or more compensators include one or more of a fluid compensator, a mechanical compensator, a magnetic compensator, or a combination thereof, wherein the one or more compensators are configured to actuate in one or more stages.

[0075] The system of any preceding clause, including the subsea structure including a subsea tree, a subsea manifold, or a combination thereof.

[0076] A method including landing a retrievable module on a subsea structure via a landing system, including moving one or more compensators of a landing post in response to a landing force, and adjusting a position of the landing post relative to a landing side of the retrievable module via a landing adjuster.

[0077] As used herein, the terms “inner” and “outer”; “up” and “down”; “upper” and “lower”; “upward” and “downward”; “above” and “below”; “inward” and “outward”; and other like terms as used herein refer to relative positions to one another and are not intended to denote a particular direction or spatial orientation. The terms “couple,” “coupled,” “connect,” “connection,” “connected,” “in connection with,” and “connecting” refer to “in direct connection with” or “in connection with via one or more intermediate elements or members.”

[0078] The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. However, the illustrative discussions above are not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. Moreover, the order in which the elements of the methods described herein are illustrated and described may be re-arranged, and/or two or more elements may occur simultaneously. The embodiments were chosen and described to best explain the principals of the disclosure and its practical applications, to thereby enable others skilled in the art to best utilize the disclosure and various embodiments with various modifications as are suited to the particular use contemplated.

[0079] Finally, the techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as “means for [perform]ing [a function] . . .” or “step for [perform]ing [a function] . . .,” it is intended that such elements are

to be interpreted under 35 U.S.C. § 112(f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. § 112(f).

Claims

1. A system, comprising: a landing system configured to land a retrievable module on a subsea structure, the landing system comprising: an outer sleeve disposed about an inner sleeve; a landing post; and a landing adjuster coupled to an axial end of the inner sleeve and configured to adjust a position of the landing post relative to a landing side of the retrievable module.
2. The system of claim 1, wherein the landing post comprises one or more compensators configured to move in response to a landing force.
3. The system of claim 2, wherein the one or more compensators comprise one or more of a fluid compensator, a mechanical compensator, a magnetic compensator, or a combination thereof.
4. The system of claim 2, wherein the one or more compensators comprise a plurality of compensators configured to actuate in a plurality of stages.
5. The system of claim 4, wherein the plurality of compensators comprises a fluid compensator comprising a piston disposed in a chamber and a mechanical compensator comprising a spring.
6. The system of claim 2, comprising a plurality of landing systems configured to land the retrievable module on the subsea structure, wherein each of the plurality of landing systems comprises the landing post and the landing adjuster.
7. The system of claim 1, wherein the landing adjuster is configured to adjust the position to stabilize landing of different masses, centers of gravity, or a combination thereof, of the retrievable module.
8. The system of claim 1, wherein the landing adjuster is configured to adjust the position of the landing post relative to the landing side of the retrievable module in one or more directions relative to the landing side.
9. The system of claim 1, wherein the landing adjuster is configured to adjust the position of the landing post relative to the landing side of the retrievable module in a first direction crosswise to the landing side.
10. The system of claim 9, wherein the landing adjuster is configured to adjust the position of the landing post relative to the landing side of the retrievable module in a second direction along the landing side.
11. The system of claim 10, wherein the first direction is vertical, and the second direction is horizontal.
12. The system of claim 1, wherein the landing adjuster comprises a manual landing adjuster, an automatic landing adjuster, or a combination thereof.
13. The system of claim 1, wherein the landing adjuster comprises one or more threaded landing adjusters.
14. The system of claim 1, comprising the retrievable module including one or more of a valve, a pump, a compressor, a flowmeter, a sensor, a chemical injection system, a fluid intervention system, a subsea sampling system, a tree conversion system, a high-integrity pressure protection system, a fluid conduit, a fluid tank, an accumulator, an actuator, a choke, a controller, or any combination thereof.
15. The system of claim 14, comprising the subsea structure including a subsea tree, a subsea manifold, or a combination thereof.
16. A system, comprising: a retrievable module; and a landing system configured to land the retrievable module on a subsea structure, the landing system comprising: an outer sleeve disposed about an inner sleeve; a landing post comprising one or more compensators configured to move in response to a landing force; and a landing adjuster coupled to an axial end of the inner sleeve and

configured to adjust a position of the landing post relative to a landing side of the retrievable module.

17. The system of claim 16, wherein the landing adjuster is configured to adjust the position of the landing post relative to the landing side of the retrievable module in one or more directions relative to the landing side, wherein the landing adjuster is configured to adjust the position to stabilize landing of different masses, centers of gravity, or a combination thereof, of the retrievable module.

18. The system of claim 16, wherein the one or more compensators comprise one or more of a fluid compensator, a mechanical compensator, a magnetic compensator, or a combination thereof, wherein the one or more compensators are configured to actuate in one or more stages.

19. The system of claim 16, comprising the subsea structure including a subsea tree, a subsea manifold, or a combination thereof.

20. A method, comprising: landing a retrievable module on a subsea structure via a landing system, comprising: moving one or more compensators of a landing post in response to a landing force; adjusting a position of the landing post in a first direction relative to a landing side of the retrievable module via a landing adjuster; and adjusting the position of the landing post in a second direction relative to the landing side of the retrievable module via the landing adjuster.
